

# EXPERIMENT 3

Name : Gayathiri T

Roll no : 240701140

## SURVEY REPORT ON EMERGING TRENDS IN LOW-CODE APPLICATION DEVELOPMENT

### 1. AIM

To study emerging trends in low-code application development, compare it with traditional development, and analyze its benefits, challenges, and existing platforms.

### 2. TRADITIONAL DEVELOPMENT VS. LOW-CODE DEVELOPMENT

Traditional development involves writing code manually using programming languages. It provides greater flexibility but requires more time and programming skills.

Low-code development uses visual tools, drag-and-drop components, and minimal coding to build applications quickly.

| Traditional Development                            | Low-Code Development                          |
|----------------------------------------------------|-----------------------------------------------|
| Requires extensive coding.                         | Requires minimal coding.                      |
| Needs programming expertise.                       | Supports users with limited coding knowledge. |
| Takes more development time.                       | Enables faster development.                   |
| Offers greater customization.                      | Customization depends on the platform.        |
| Testing and deployment are often managed manually. | Provides built-in development tools.          |

Emerging Trends:

- AI-assisted application development.
- Workflow automation.
- Cloud-based development.
- Citizen development.
- Integration with APIs and existing systems.

### 3. BENEFITS AND CHALLENGES OF LOW-CODE

Benefits:

1. Faster application development.
2. Reduced coding effort.
3. Easy prototyping.

4. Improved productivity.
5. Simplified workflow automation.

Challenges:

1. Vendor dependency and lock-in.
2. Limited customization for complex applications.
3. Security and privacy concerns.
4. Possible scalability limitations.
5. Licensing and maintenance costs.

#### 4. COMPARISON OF EXISTING PLATFORMS

| Platform             | Main Features                        | Common Uses                 |
|----------------------|--------------------------------------|-----------------------------|
| Microsoft Power Apps | Visual tools and data integration    | Business applications       |
| Zoho Creator         | Forms, reports, and Deluge scripting | Business process automation |
| Mendix               | Visual development and collaboration | Enterprise applications     |
| OutSystems           | Web and mobile app development       | Enterprise solutions        |
| Appian               | Workflow and process automation      | Business process management |

#### 5. RESULT

The survey on emerging trends in low-code application development was completed successfully. The study compared traditional and low-code development, identified their benefits and challenges, and examined existing platforms. Low-code development helps create applications faster with less coding, but security, customization, and scalability must be considered.